Research Proposal: Do Geopolitical Threats Move the Fed?

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Executive Summary

- **Research question:** Does a geopolitical risk shock behave like a cost-push shock in the U.S. economy, that is, does it raise inflation while lowering output? Do geopolitical threats (GPT) and geopolitical acts (GPA) transmit differently?
- **Project logic:** This report covers only the *first stage* of the broader project. The present goal is to document the reduced-form responses of industrial production and inflation to GPR, GPT, and GPA shocks. The next stage will examine the underlying channels, such as oil prices, raw materials, expected inflation, and credit spreads.
- **Why separate GPR, GPT, and GPA?** The headline GPR index may combine different mechanisms. GPT is more naturally associated with expectations, uncertainty, and risk premia, while GPA is more naturally associated with realized disruptions, including commodity and supply-chain effects.
- **Data:** The analysis uses monthly U.S. data for 1990–2025. The main outcome variables are industrial production growth and inflation. Robustness checks additionally include the unemployment rate and VIX growth.
- **Methodologies:** Two methods are used: ADL and Local Projections (LP). ADL is used as a benchmark. LP is the main method for the broader proposal. In both approaches, impulse responses are scaled to a one-standard-deviation shock.
- **Descriptive evidence:** GPR, GPT, and GPA are highly correlated with each other, but their contemporaneous correlations with macro-financial variables are modest. This suggests that the effects of geopolitical risk are likely to be dynamic rather than purely contemporaneous.

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- **Results of ADL:** ADL tends to imply a negative response of industrial production across the three shock measures. Inflation responses are mixed: GPR and GPT are mildly positive at some horizons, while GPA tends to be weaker or negative.
- **Results of LP:** Local Projections implie a positive output response and a mostly negative or near-zero inflation response. This weakens the reduced-form case for a clean cost-push interpretation.
- **Robustness Checks:** Within each method, adding the unemployment rate and VIX growth changes the impulse responses only modestly. The main sensitivity in the results comes from the choice of estimation method rather than from the inclusion of these controls.

- **Main takeaway:** The current reduced-form evidence does not yet deliver one clean confirmation of the cost-push hypothesis. Instead, it motivates the next stage of the project: channel decomposition, especially a comparison of how GPT and GPA propagate through different mechanisms.

1 Introduction

This report presents the first empirical step of a broader project on whether geopolitical risk shocks behave like cost-push disturbances in the U.S. economy. The central question is whether a positive geopolitical risk shock raises inflation while lowering output, and whether this transmission differs between geopolitical threats (GPT) and geopolitical acts (GPA). The broader motivation is to begin with the reduced-form responses of industrial production and inflation, and then move to a decomposition of the underlying channels, including oil prices, raw materials, expected inflation, and credit spreads.

The logic of this first stage is straightforward. Before identifying specific transmission channels, it is necessary to establish whether the aggregate reduced-form responses display the basic cost-push signature. If geopolitical risk shocks do not generate a pattern consistent with higher inflation and lower output even at the reduced-form level, then a channel-based interpretation becomes less compelling. Conversely, if such a pattern is present, the next step is to ask which mechanism is dominant and whether the Federal Reserve's response can be interpreted in light of that transmission.

A further motivation for this exercise is that the headline GPR index may conceal important heterogeneity across subcomponents. Caldara and Iacoviello's decomposition into GPT and GPA suggests that threats and acts may propagate through different mechanisms. GPT is more naturally associated with expectations, uncertainty, and risk premia, whereas GPA is more naturally associated with realized disruptions, including commodity-price and supply-side effects. For this reason, the empirical analysis is carried out separately for GPR, GPT, and GPA rather than relying only on the aggregate index.

The empirical strategy uses two methods. ADL is employed as a benchmark dynamic specification, while Local Projections (LP) are treated as the main methodology for the broader project. LP is particularly attractive in this context because it delivers horizon-by-horizon impulse responses and is generally more robust to dynamic misspecification. The current exercise therefore uses ADL to provide a structured benchmark and LP to provide the primary reduced-form evidence for the proposal.

The results of this first-stage exercise are informative but not decisive. The descriptive evidence confirms that GPR, GPT, and GPA are highly correlated but not identical, and that major geopolitical episodes generate clear spikes in all three series. Within each method, adding unemployment and VIX controls leaves the estimated responses largely unchanged. Across methods, however, the conclusions differ substantially: ADL tends to imply a negative industrial production response and a mixed inflation response, whereas LP implies a positive output response and mostly negative or near-zero inflation responses. The main implication is therefore not that the cost-push hypothesis is already confirmed, but that the reduced-form evidence motivates the next stage of the project, namely a channel decomposition that can distinguish more clearly between threats, acts, and their underlying transmission mechanisms. In this sense, this report shows a reduced-form diagnostic for the broader project rather than as the final structural interpretation of geopolitical risk shocks.

2 Data and Variable Construction

The analysis uses monthly U.S. data from 1990 to 2025. The three geopolitical risk measures are the headline geopolitical risk index (GPR), geopolitical threats (GPT), and geopolitical acts (GPA). For descriptive purposes, I plot the raw indices in order to preserve the historical scale and to highlight major geopolitical episodes directly. For estimation, however, the shock variables are transformed into normalized log levels:

$$\begin{aligned} gpr_n_t &= 100 \log(100 \cdot GPR_t), \\ gpt_n_t &= 100 \log(100 \cdot GPT_t), \\ gpa_n_t &= 100 \log(100 \cdot GPA_t) \end{aligned} \tag{1}$$

This transformation serves two purposes. First, the log specification makes the shock measures easier to interpret in proportional terms rather than in raw index units. Second, the additional scaling by 100 places the variables in a numerically convenient range, which makes the estimated impulse responses easier to read and compare across specifications. In this sense, the transformed GPR variables are intended as estimation-ready shock measures, whereas the raw indices are retained only for historical and descriptive visualization.

The macroeconomic variables are constructed as follows:

$$dlog_ip_t = 100\, \Delta \log(INDPRO_t), \tag{2}$$

$$infl_t = 100\, \Delta \log(CPI_t), \tag{3}$$

$$dlog_wti_t = 100\, \Delta \log(WTI_t), \tag{4}$$

$$dlog_vix_t = 100\, \Delta \log(VIX_t). \tag{5}$$

The federal funds rate and unemployment rate are kept in levels.

These transformations reflect the role played by each variable in the empirical exercise. Industrial production and CPI are expressed in log differences because the first-stage objective is to study output and inflation responses rather than the behavior of their levels. By contrast, the policy rate and unemployment rate are retained in levels because they are used as control variables in the robustness exercises. WTI growth and VIX growth are included both as descriptive macro-financial indicators and as variables that become directly relevant for the subsequent channel analysis.

Because the transformed outcome variables rely on first differences, the effective sample for growth variables is one observation shorter than for level variables. In addition, the exact usable sample may vary slightly across specifications once additional controls are included. These differences are small, but they should be kept in mind when comparing the number of usable observations across tables and regressions. Industrial production and CPI are expressed in log differences because the first-stage exercise focuses on output growth and inflation dynamics. In the dynamic analysis below, these transformed variables are used as estimation inputs, while the reported impulse responses are interpreted cumulatively under the relevant ADL and LP setups.

3 Descriptive Evidence

3.1 Summary Statistics

Table 1 reports summary statistics for the transformed variables. The three geopolitical risk measures have broadly similar means and standard deviations, reflecting their strong common component, although GPA is somewhat more volatile than GPR and GPT. Among the macro-financial variables, VIX growth exhibits by far the largest volatility, while industrial production growth also displays substantial tail behavior. The higher moments indicate strong non-normality in several series, especially IP growth and VIX growth, reflecting the importance of crisis episodes in the sample. This feature is relevant for the empirical analysis below, as it suggests that estimated dynamic responses may be sensitive to the treatment of persistence, volatility, and tail events.

Table 1: Summary Statistics

Variable	Mean	Std	Min	Max	Skewness	Kurtosis	N
GPR	916.84	35.959	826.99	1084.5	0.9925	5.9147	432
GPT	920.05	36.264	820.76	1060.6	0.7072	4.3862	432
GPA	907.84	48.423	795.35	1135.5	0.9775	5.5379	432
IP growth	0.1158	1.0296	-14.142	6.3472	-6.0670	91.702	431
Inflation	0.2183	0.2685	-1.7864	1.3675	-0.9094	12.654	429
FFR	2.8793	2.3432	0.05	8.29	0.2492	1.7875	432
WTI growth	0.2159	9.5608	-56.819	54.585	-0.6151	10.879	431
VIX growth	-0.0943	16.274	-44.61	107.89	1.4233	9.1152	431
Unrate	5.6773	1.7417	3.4	14.8	1.2735	5.0956	431

3.2 Historical Behavior of GPR, GPT, and GPA

Figure 1 plots the raw GPR, GPT, and GPA indices. All three series display pronounced spikes around major geopolitical episodes, including the Gulf War, the September 11 attacks, the Iraq War, and the Russia–Ukraine conflict. At the same time, the subcomponents exhibit visibly different dynamics. GPA tends to generate sharper and shorter-lived spikes, whereas GPT appears smoother and more persistent. This distinction is consistent with the interpretation of GPA as realized geopolitical events and GPT as forward-looking threat news. It also provides a direct motivation for estimating the full empirical model separately for GPR, GPT, and GPA rather than relying only on the aggregate headline index.

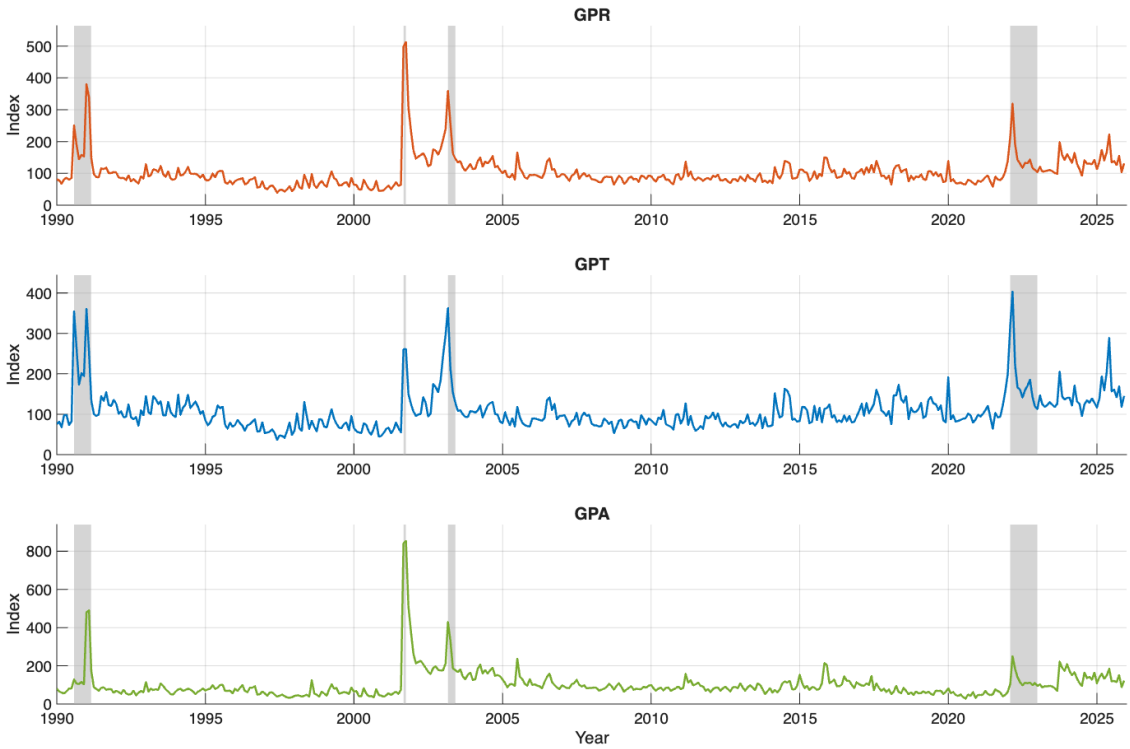

Figure 1: GPR, GPT, and GPA Indices

3.3 Macro-financial Variables

Figure 2 presents the transformed macro-financial variables. The figure highlights substantial volatility in industrial production growth, oil-price growth, and especially VIX growth, with particularly large swings during crisis periods. Inflation is comparatively more stable, though it also shows visible deviations in high-volatility episodes. These patterns are important for the empirical analysis because they suggest that the estimated responses of output and inflation may depend on how the dynamic specification handles persistence, crisis-related outliers, and volatility clustering.

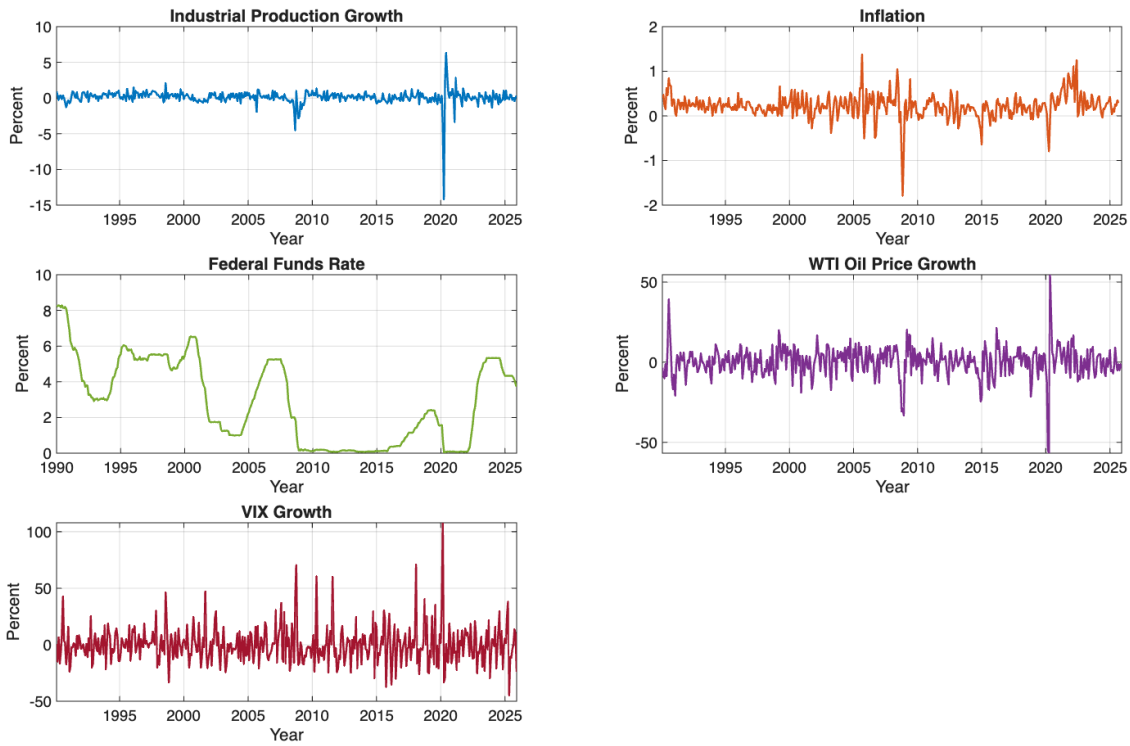

Figure 2: Macro-financial Variables

3.4 Correlation Structure and Persistence

Figure 3 reports the correlation matrix. As expected, the three geopolitical risk measures are highly correlated with one another, confirming that they capture a shared geopolitical-risk component. At the same time, their correlations with macro-financial variables are generally modest in contemporaneous terms. Inflation and oil-price growth are positively correlated, while the federal funds rate is negatively correlated with unemployment. The main implication is that geopolitical risk is unlikely to operate primarily through strong same-period comovement; instead, its effects are more likely to emerge through dynamic propagation.

Figure 4 reports the autocorrelation functions of GPR, GPT, and GPA. All three series are highly persistent, though the precise degree of persistence differs across measures. This persistence provides an additional motivation for the dynamic methods used below and further supports treating GPT and GPA separately in the empirical analysis.

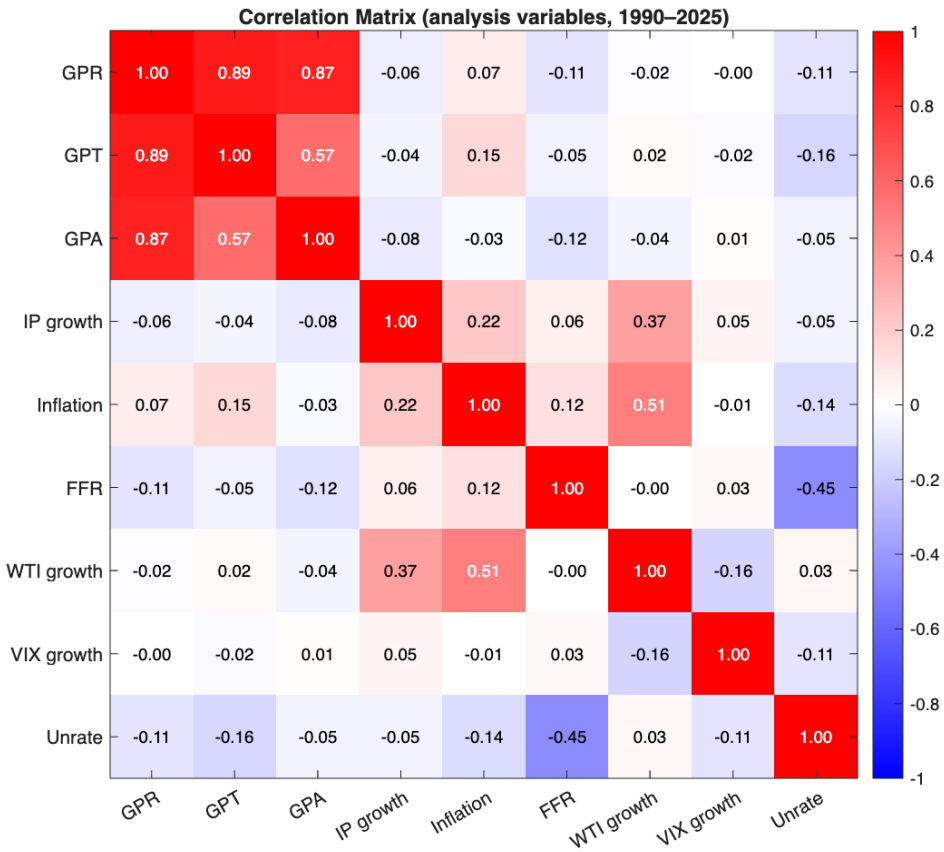

Figure 3: Correlation Matrix

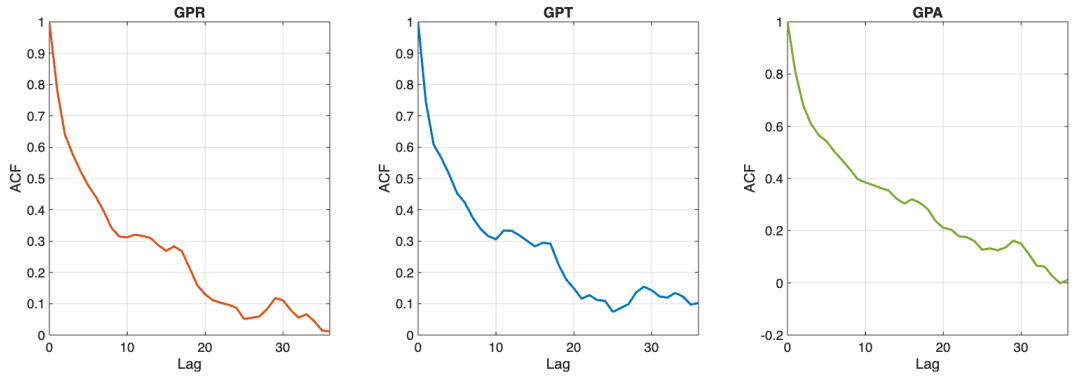

Figure 4: Autocorrelation Functions of GPR, GPT, and GPA

4 Empirical Methodology

4.1 Stage 1: Reduced-form responses of output and inflation

The current exercise corresponds to the first stage of the broader project. At this stage, the objective is not yet to identify individual transmission channels, but to determine whether geopo-

litical risk shocks generate the basic reduced-form pattern associated with a cost-push disturbance. In practice, this means asking whether a positive GPR-type shock is followed by lower industrial production and higher inflation. This first-stage exercise is useful even if the resulting evidence is mixed, because it disciplines the next step of the project: channel decomposition. If the reduced-form responses already differ across GPR, GPT, and GPA, then the subsequent channel analysis becomes more informative and more necessary.

4.2 ADL

As an initial benchmark, I estimate single-equation autoregressive distributed lag (ADL) models using the transformed geopolitical risk measures and the two outcome variables. The ADL specification is useful because it imposes a structured lag representation and therefore provides a disciplined summary of the dynamic relationship between shocks and outcomes. In this sense, ADL serves as a benchmark against which the Local Projection results can later be compared.

For the ADL specification, the horizon is set to $H = 36$, the shock lag length is set equal to the horizon, and the number of endogenous lags is selected from 1 to 12 using BIC. The outcomes are industrial production growth and inflation, and impulse responses are reported for a one-standard-deviation shock.

4.3 Local Projections

The main estimation method for the broader project is Local Projections (LP). LP is particularly well suited to this setting because it estimates horizon-specific responses directly, without imposing a tight dynamic structure across all horizons. This is useful in the present context, where the propagation of geopolitical risk may differ across GPR, GPT, and GPA, and where later stages of the project will focus on transmission channels and potentially heterogeneous dynamics. Relative to more structured specifications, LP therefore provides a flexible reduced-form framework for tracing the dynamic effects of geopolitical shocks.

The LP specification uses horizon $H = 36$, $I = 12$ endogenous lags, a contemporaneous shock only ($J = 0$), and a linear time trend. Confidence intervals are based on Newey–West standard errors using the default rule-of-thumb bandwidth. As in the ADL case, the LP estimates are scaled to represent a one-standard-deviation shock.

4.4 Baseline and robustness specifications

For both ADL and LP, three specifications are estimated:

1. Baseline: shock only.
2. Robustness 1: shock + unemployment rate.
3. Robustness 2: shock + unemployment rate + VIX growth.

These robustness checks are not intended to represent the full channel decomposition. Rather, they provide a preliminary assessment of whether the reduced-form responses are sensitive to the inclusion of additional macro-financial controls. This distinction is important. Variables such as oil prices, raw materials, expected inflation, and credit spreads are conceptually part of the channel analysis that belongs to the next stage of the project. By contrast, unemployment and VIX are used here only as broad controls to assess whether the first-stage reduced-form responses are robust to labor-market and financial-market conditions.

4.5 Interpretation within the broader project

The broader research design is organized around the following logic. If GPR shocks behave like cost-push disturbances in the first-stage reduced-form estimates, the next step is to decompose the responses into specific channels such as oil prices, raw materials, expected inflation, and credit spreads. The GPT versus GPA split is especially important because threats may operate mainly through expectations and risk premia, whereas acts may operate more directly through physical supply disruption. The current section should therefore be interpreted as a reduced-form diagnostic for the broader project rather than as the final structural answer to the full research question.

5 Results

5.1 ADL Results

Figure 5 reports the baseline ADL impulse responses. The clearest feature is a negative industrial production response across all three geopolitical shock measures. The contraction is particularly visible at medium horizons and is most pronounced for GPT and GPA. By contrast, the inflation responses are less uniform. GPR and GPT are associated with mildly positive responses at some horizons, whereas GPA tends to display a weaker and in some cases negative inflation response. Taken together, the ADL estimates provide only partial support for a cost-push interpretation: the output response is broadly consistent with contractionary geopolitical shocks, but the inflation response is not uniformly positive across shock definitions.

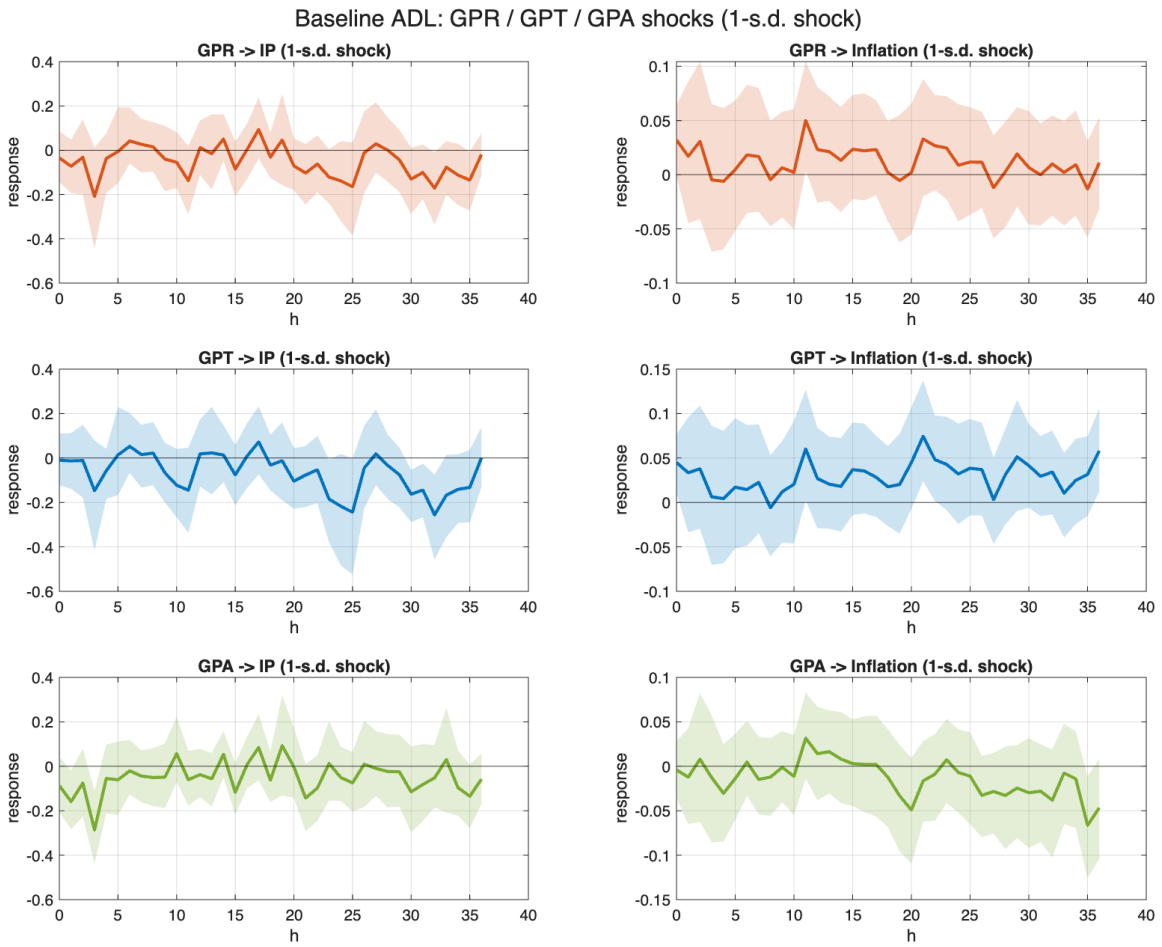

Figure 5: Baseline ADL Impulse Responses

Figure 6 and Figure 7 report the robustness checks. Adding unemployment and then VIX growth leaves the response paths largely unchanged. In particular, the negative industrial production response remains visible across all three shock measures, while the inflation responses continue to differ mainly across GPR, GPT, and GPA rather than across specifications. This suggests that, within the ADL framework, the main qualitative results are not being driven by omitted labor-market or financial-market conditions.

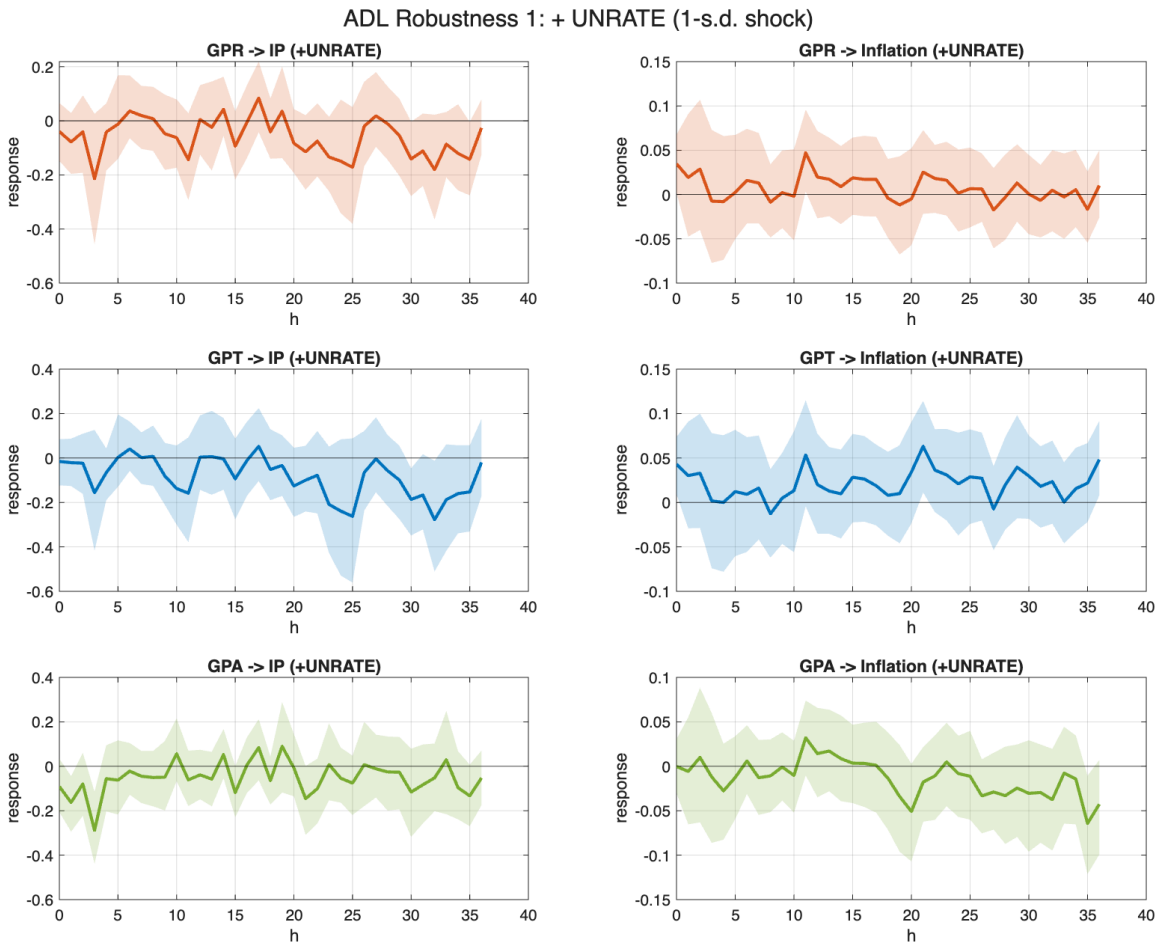

Figure 6: ADL Robustness 1: + UNRATE

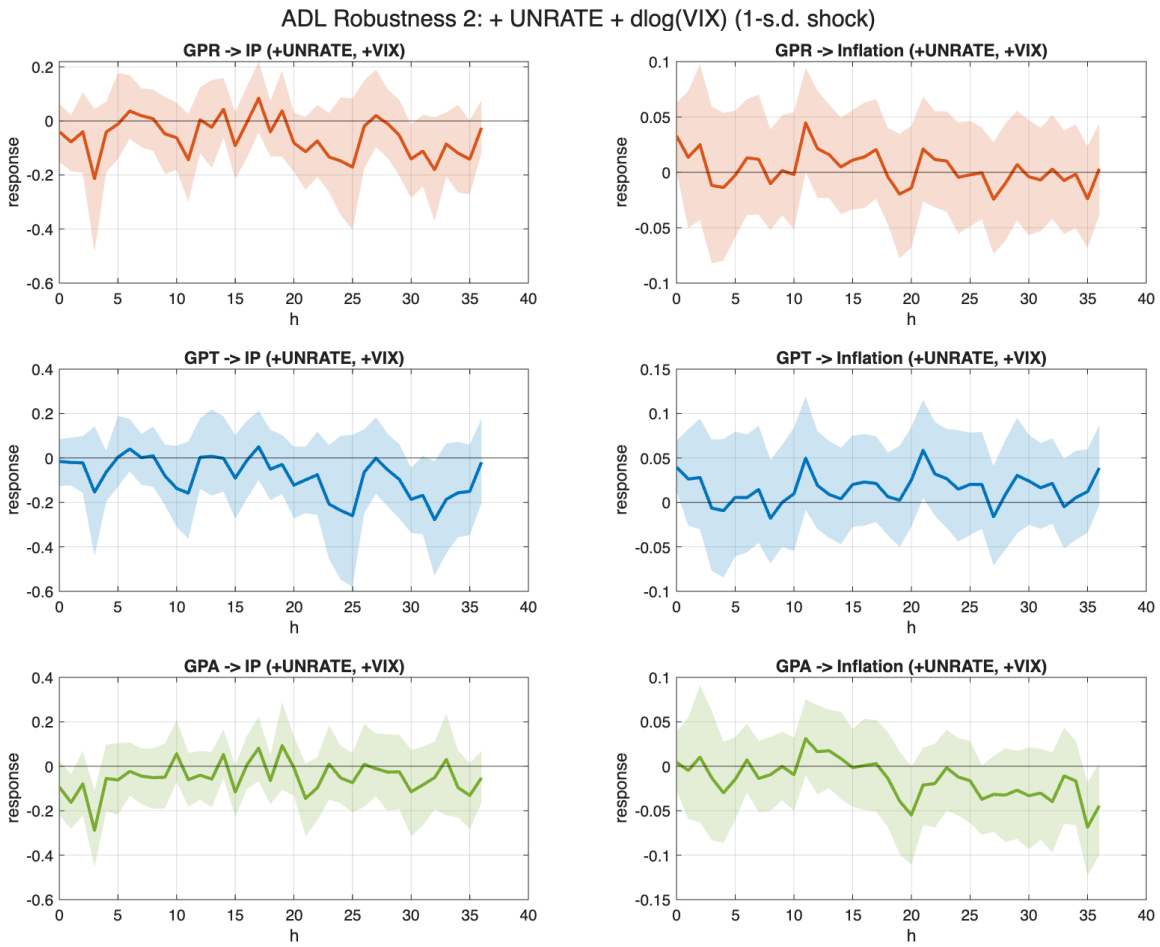

Figure 7: ADL Robustness 2: + UNRATE + dlog(VIX)

Figure 8 overlays the three ADL specifications. The response paths lie very close to one another, confirming that the ADL estimates are highly stable across the robustness exercises. This stability strengthens the interpretation of the ADL results as a coherent within-method benchmark: the main empirical variation comes from the choice of shock decomposition rather than from the inclusion of unemployment or VIX controls.

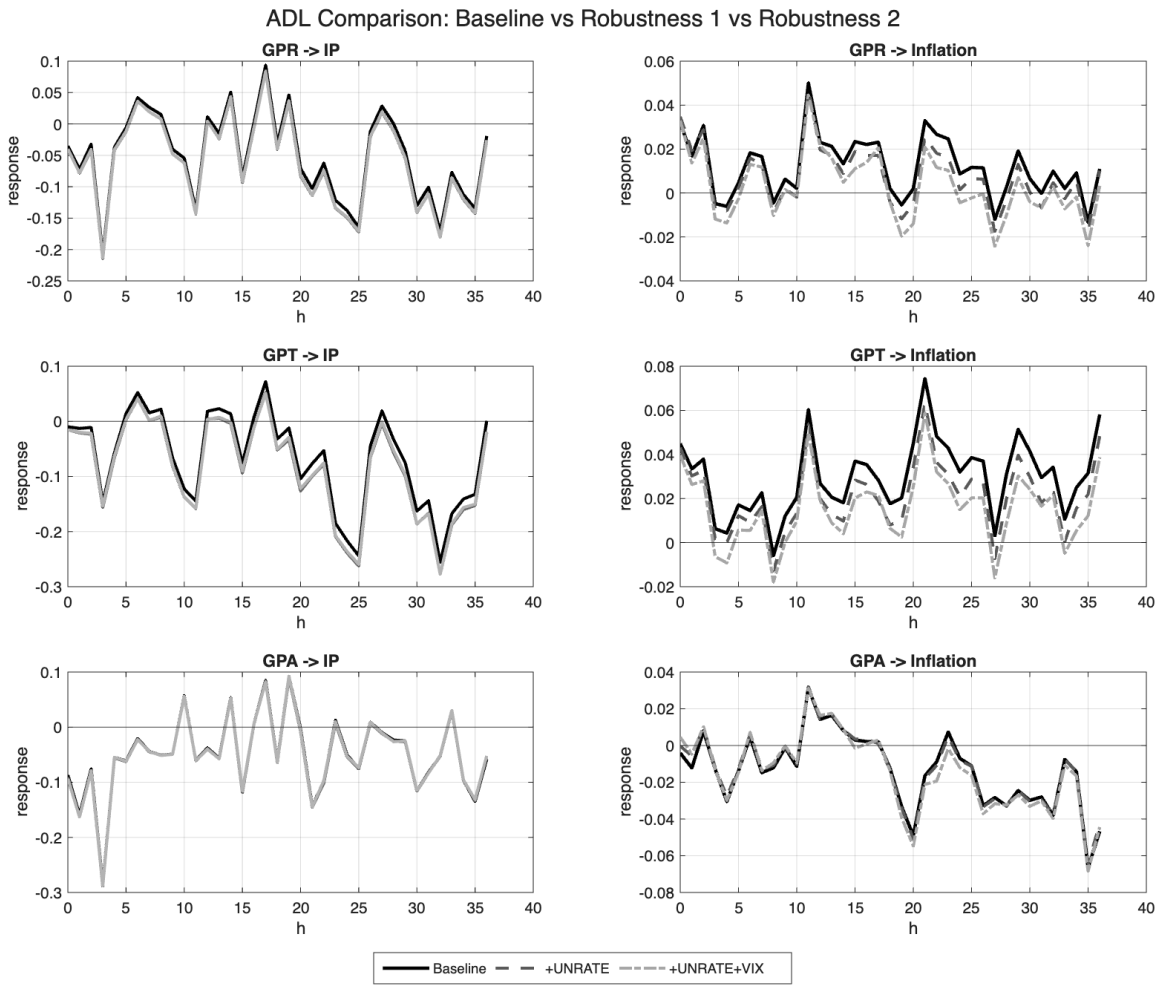

Figure 8: ADL Comparison across Specifications

5.2 LP Results

Figure 9 presents the baseline LP impulse responses. In contrast to the ADL results, the LP estimates imply a positive response of industrial production for all three geopolitical risk measures. The inflation responses are generally negative or close to zero. Taken at face value, these reduced-form LP estimates do not support a cost-push interpretation of geopolitical risk shocks. More importantly, the contrast with the ADL evidence already indicates that the empirical conclusions are sensitive to the dynamic specification used to recover the impulse responses.

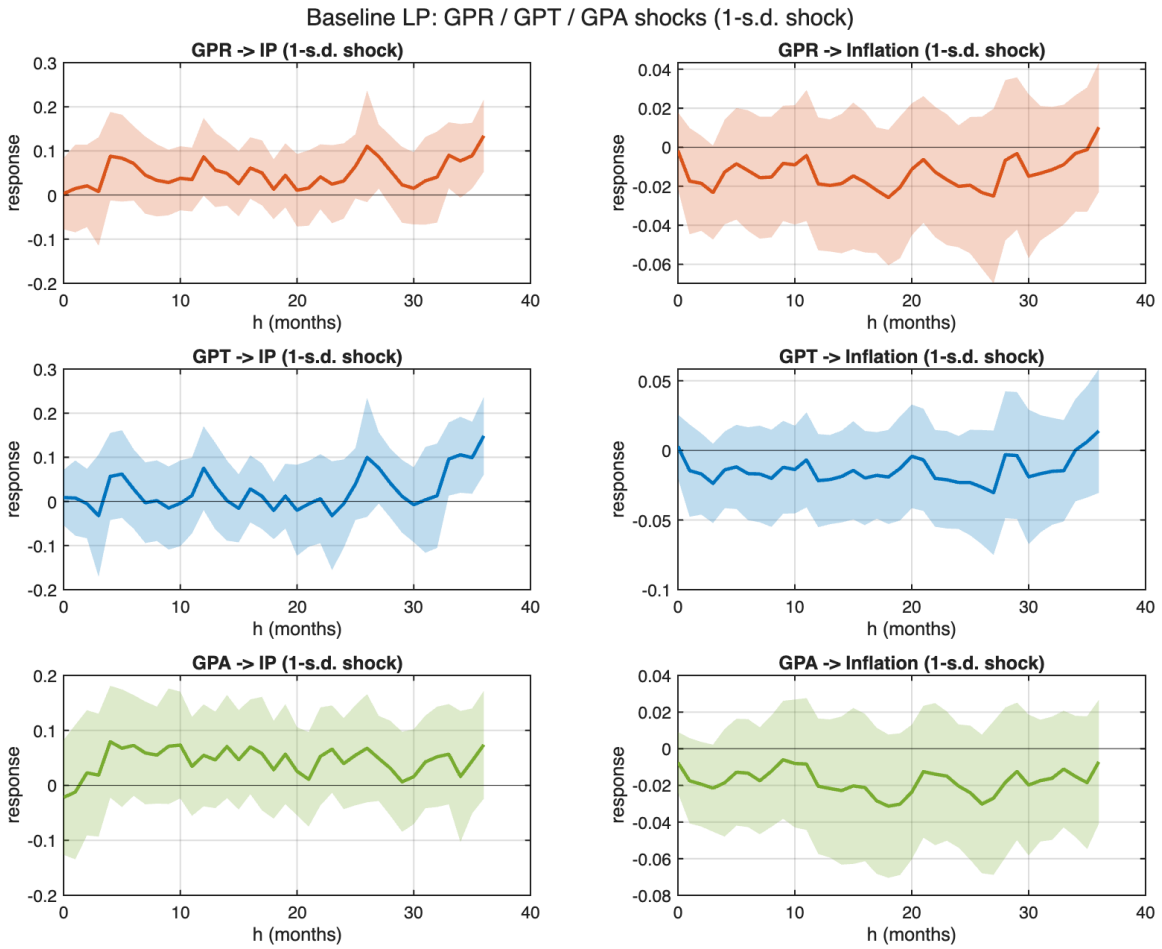

Figure 9: Baseline LP Impulse Responses

Figure 10 and Figure 11 report the LP robustness checks. As in the ADL case, the inclusion of unemployment and VIX growth does not materially alter the estimated paths. The positive output response and the mostly negative or near-zero inflation response remain intact across all three specifications. This suggests that, within the LP framework, the reduced-form findings are highly robust to the addition of these macro-financial controls.

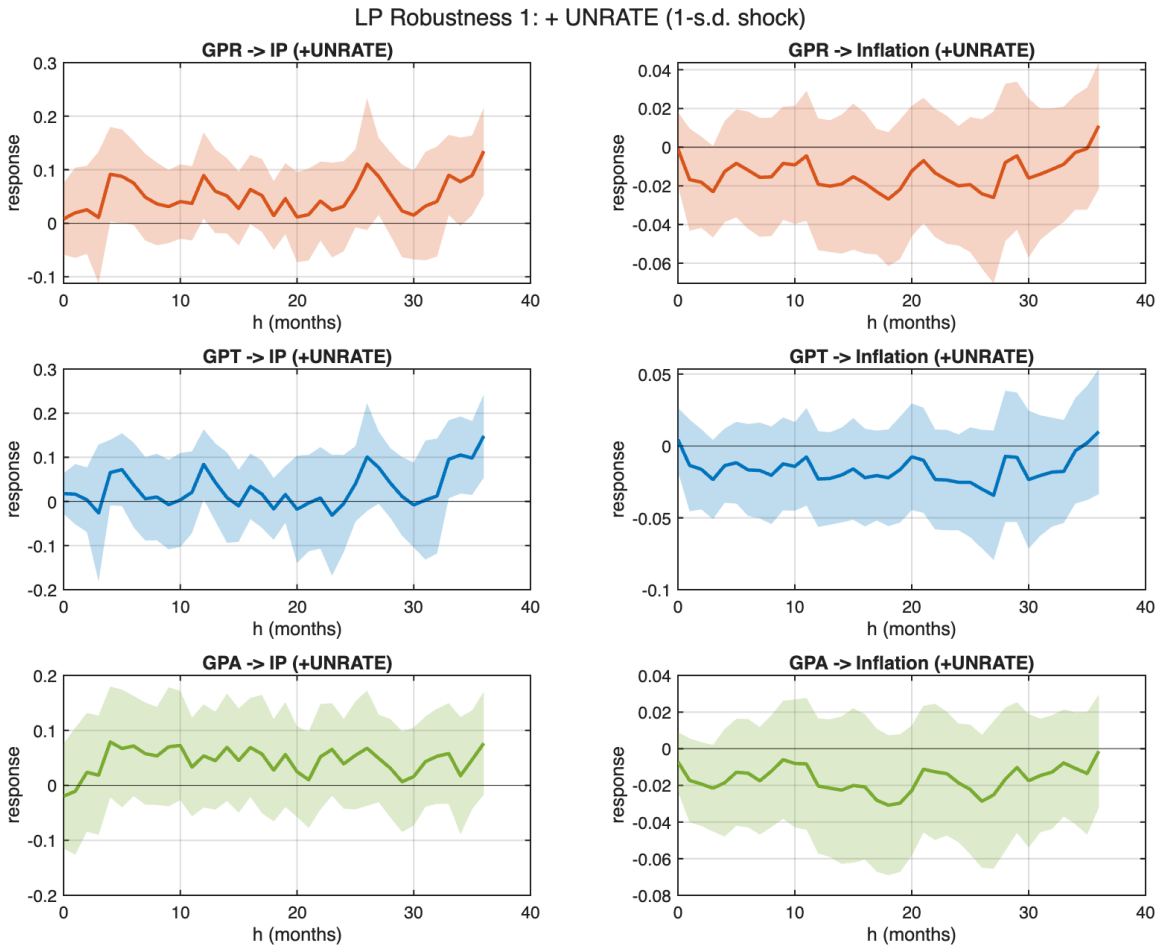

Figure 10: LP Robustness 1: + UNRATE

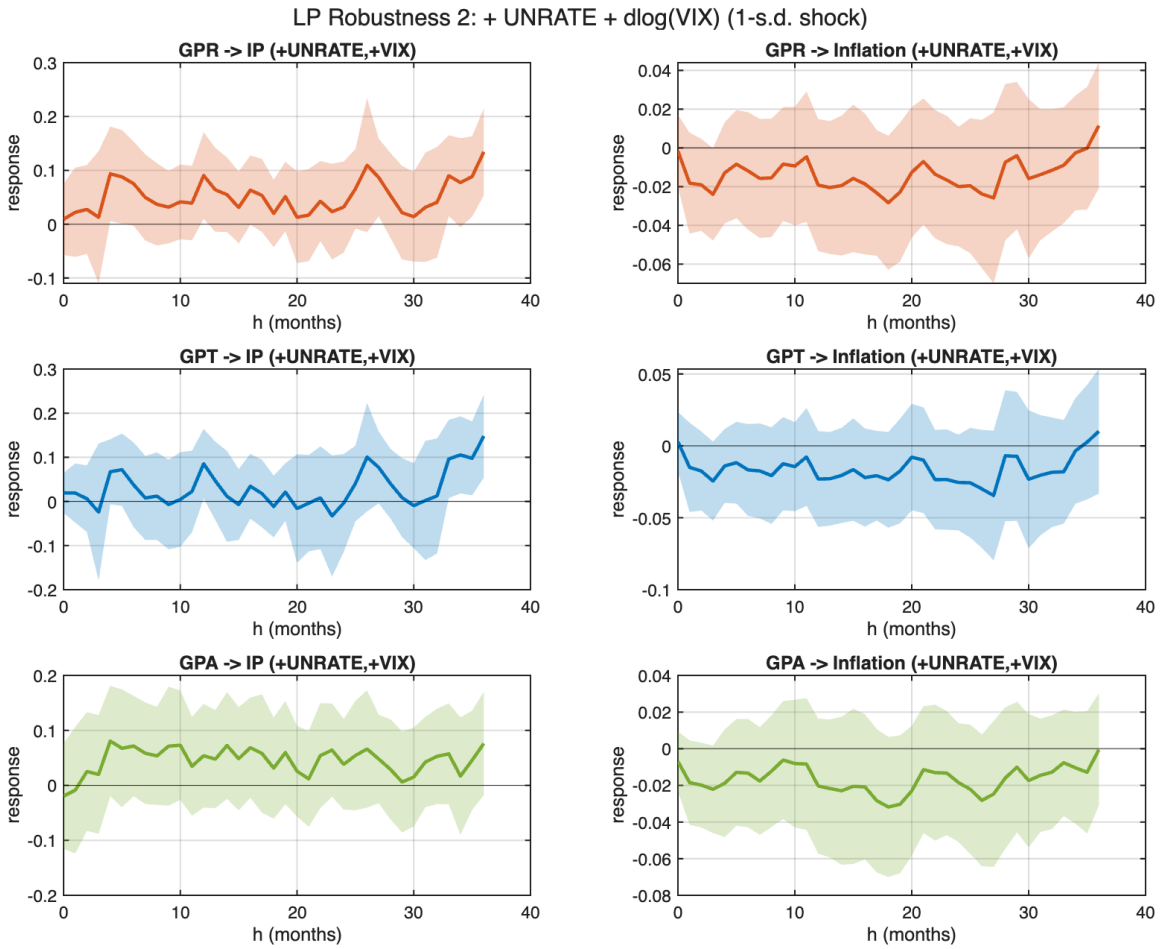

Figure 11: LP Robustness 2: + UNRATE + dlog(VIX)

Figure 12 overlays the LP specifications. The three lines are nearly indistinguishable in all panels, reinforcing the conclusion that the LP results are remarkably stable within method. The main source of empirical disagreement is therefore not the inclusion of unemployment or VIX, but the contrast between LP and ADL as alternative dynamic estimators.

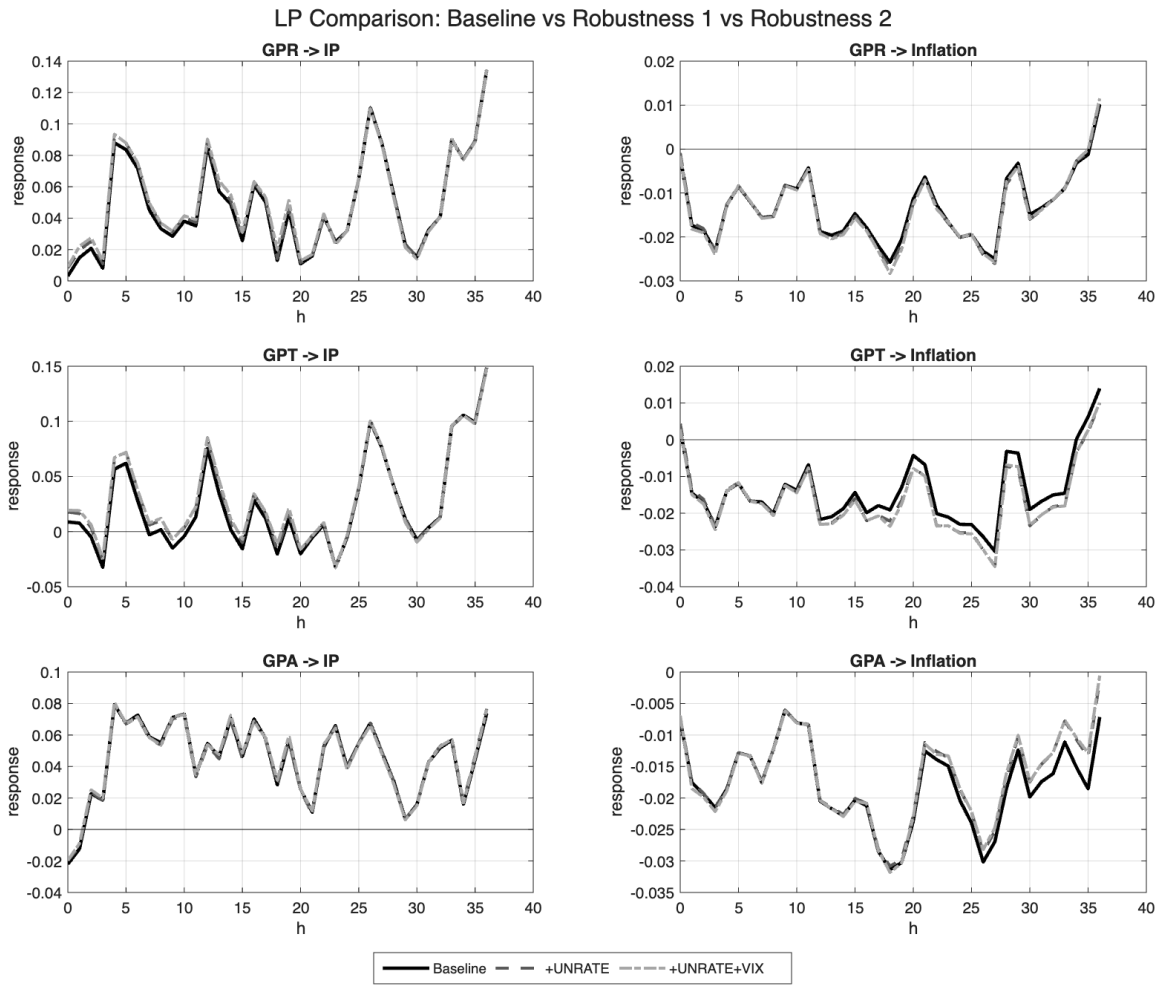

Figure 12: LP Comparison across Specifications

5.3 ADL versus LP

Figure 13 and Figure 14 compare the baseline ADL and LP responses directly. The contrast is substantial. For industrial production, ADL implies a negative response, whereas LP implies a positive response. For inflation, ADL tends to generate mildly positive responses for GPR and GPT, while LP produces responses that are mostly negative or close to zero. These differences are economically meaningful and cannot be dismissed as minor specification noise.

The key implication is that the first-stage reduced-form evidence is much more sensitive to the choice of dynamic estimator than to the inclusion of unemployment or VIX controls. In other words, the main empirical uncertainty does not come from the robustness exercises within each method, but from the fact that ADL and LP summarize the underlying dynamics differently. This is precisely why the present stage should be interpreted as diagnostic rather than conclusive: the reduced-form responses are informative, but they do not yet pin down a single structural interpretation of geopolitical risk shocks.

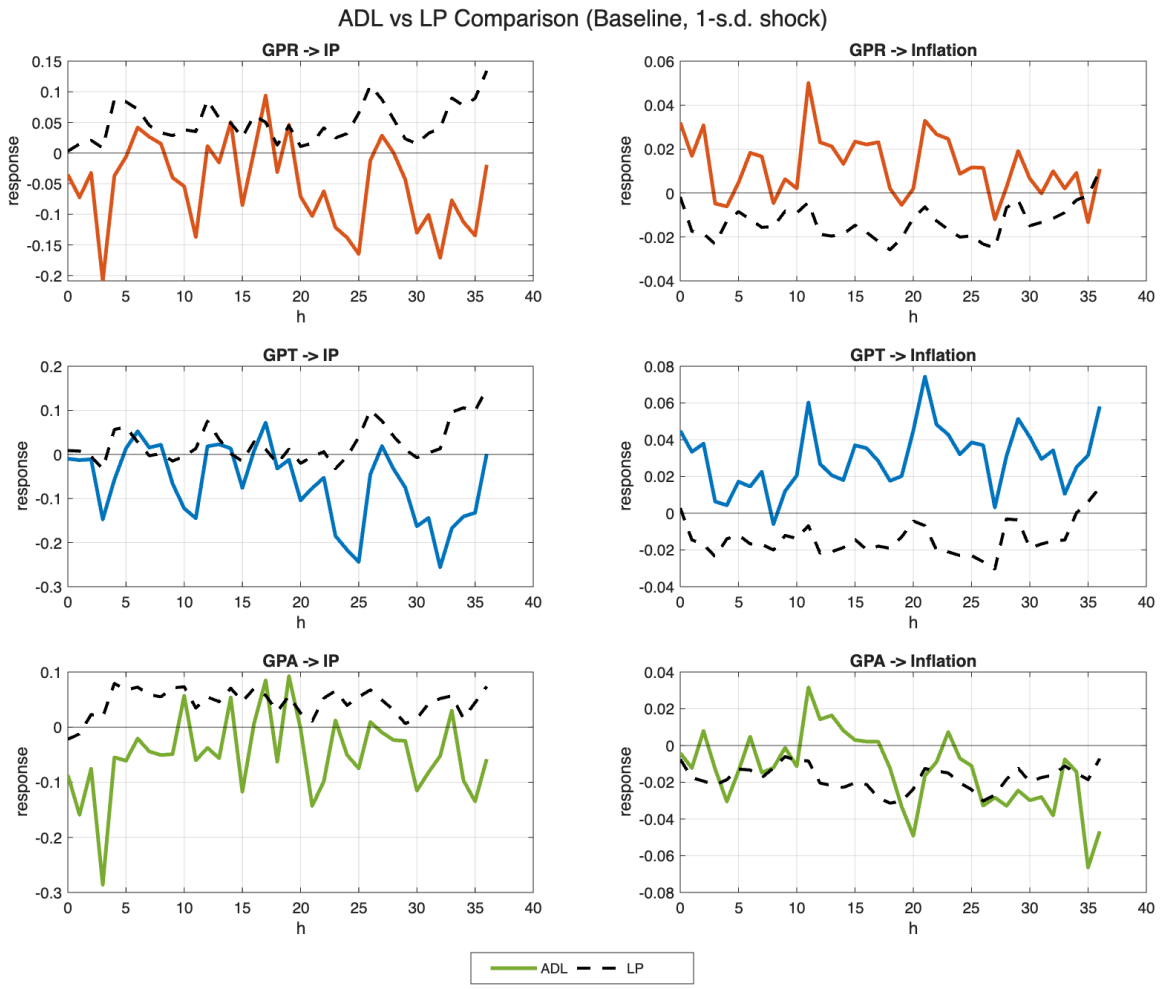

Figure 13: ADL versus LP: Baseline Comparison

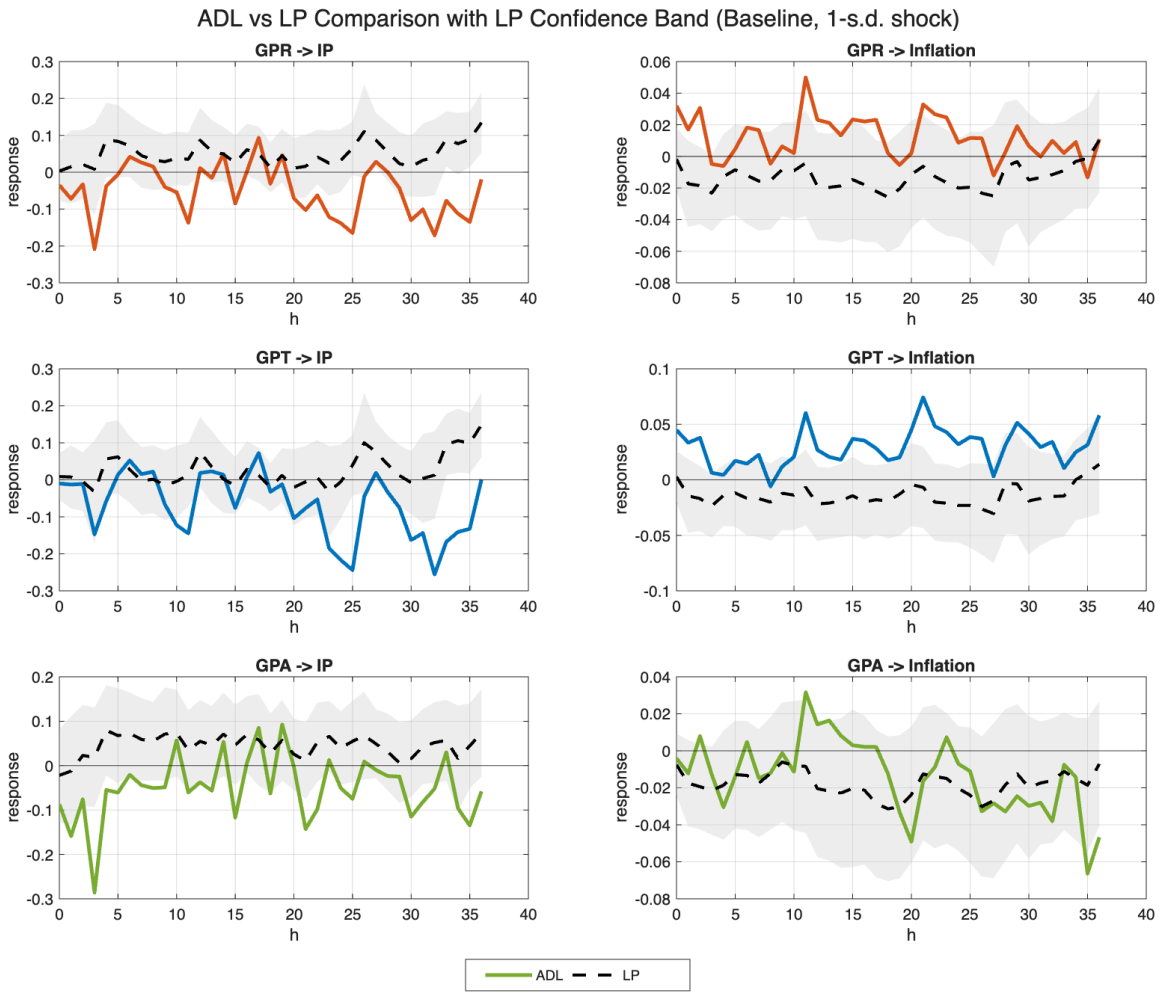

Figure 14: ADL versus LP with LP Confidence Bands

6 Interpretation

The most robust finding in this first-stage exercise is that the estimated impulse responses are highly stable within method. In both ADL and LP, adding unemployment and then VIX growth changes the response paths only modestly. This suggests that the basic reduced-form patterns are not being driven primarily by these additional macro-financial controls. Put differently, the main qualitative responses are not especially sensitive to the inclusion of broad labor-market and financial-market controls.

What is not robust, however, is the comparison across methods. The ADL estimates are more consistent with the output side of a cost-push interpretation, since industrial production tends to decline after a GPR-type shock. At the same time, the inflation response remains mixed and depends on the shock decomposition, with GPR and GPT showing mildly positive responses at some horizons and GPA showing weaker or more negative responses. The LP estimates point in a different direction: output responses are generally positive, while inflation responses are mostly

negative or near zero. The main empirical tension in this report therefore lies not in the robustness checks within each method, but in the fact that ADL and LP summarize the underlying dynamics differently.

This distinction is central for the broader project. The present stage is designed as a reduced-form diagnostic, not as a final structural interpretation of geopolitical risk shocks. The fact that GPR, GPT, and GPA already produce different reduced-form patterns suggests that the aggregate headline index may conceal important heterogeneity in transmission. If threats operate mainly through expectations and risk premia while acts operate more directly through commodity-market and supply-side disruptions, then the net reduced-form responses may reflect offsetting mechanisms rather than a single dominant effect. For this reason, the current results motivate, rather than replace, the next stage of the project: channel decomposition, with particular emphasis on comparing the dominant transmission mechanisms of GPT and GPA. This is especially relevant for the GPT–GPA decomposition, since the present reduced-form results already suggest that threats and acts may differ not only in persistence but also in the sign and timing of their inflation responses.

7 Conclusion

This report presented the first-stage reduced-form evidence for a broader project on whether geopolitical risk shocks behave like cost-push disturbances in the U.S. economy. The immediate objective was to examine whether GPR, GPT, and GPA shocks generate the basic pattern of lower output and higher inflation, before moving to a more detailed analysis of the underlying transmission channels.

The main empirical conclusion is mixed but informative. Within each method, the estimated responses are highly stable to the inclusion of unemployment and VIX controls. Across methods, however, the results differ substantially. ADL tends to imply a negative output response and a mixed inflation response, whereas LP implies a positive output response and mostly negative or near-zero inflation responses. The first-stage evidence therefore does not deliver a single clean confirmation of the cost-push hypothesis.

At the same time, these results clarify the agenda for the next stage of the project. The fact that GPR, GPT, and GPA already display different reduced-form patterns suggests that the aggregate headline index may conceal important heterogeneity in transmission. The natural next step is therefore to move to a channel decomposition that examines the role of oil prices, raw materials, expected inflation, credit spreads, and ultimately the policy response of the Federal Reserve. In that sense, the present report should be viewed not as the final answer, but as the empirical foundation for the next stage of the analysis. More broadly, the current findings suggest that any interpretation of the Federal Reserve's response to geopolitical shocks should be grounded in a prior understanding of which transmission channel is dominant and whether that dominance differs across threats and acts.